Knowledge Integration in Task-Tuned Financial LLMs

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1 Problem

Leaving Large Language Models (LLMs) static post-deployment leads to performance degradation as the world changes [3]. While fine-tuning can update a model's knowledge, it risks overwriting previously learned capabilities [5]. Several knowledge update strategies have been proposed such as locate-and-edit methods such as AlphaEdit [2] and continual alignment methods such as COPR [7]. However, they have been evaluated almost exclusively on base or instruction-tuned models [4].

However, deployed models are often fine-tuned for specific downstream tasks. In several cases, these models must know both facts relevant to its task and the necessary skills to complete the task. In the case where one needs to update an fine-tuned LLM's knowledge, one might look towards weight-space editing methods. However, a recent study found that fine-tuning moves activations in directions nearly orthogonal to knowledge editing directions, and that AlphaEdit edits are more fragile than MEMIT edits under subsequent fine-tuning [6]. This suggestions a potential conflict with methods such as AlphaEdit. A fine tuned model might encode learned skills as distributional strategies over a larger input space, however methods such as AlphaEdit only guarantee point wise preservation on a finite set of inputs.

In this paper, we want to investigate whether using policy preservation techniques (regularizing the model's output distribution rather than constraining individual weight updates) better preserves task performance in on models fined tuned for a specific task. We study this in the financial domain. Our primary setting is a model fine-tuned to decompose financial queries into sub-queries for retrieval-augmented generation (RAG).

2 Contributions

1. **Adapting COPR for factual knowledge injection:** We adapt COPR [7] for factual knowledge updates by using factual correctness rankings instead of the original human preference rankings used in the paper. For each knowledge update, we 1) sample multiple model responses, 2) rank them by ground-truth accuracy, and 3) apply COPR's advantage-weighted policy fitting with task replay regularization.

2. **Comparison of knowledge update methods:** We compare weight-space editing (AlphaEdit), policy-space regularization (COPR-adapted), and SFT baselines on task-tuned models.

3 Methods

3.1 Weight-space preservation (AlphaEdit)

AlphaEdit [2] edits a single FFN weight matrix W by projecting a perturbation Δ onto the null space of preserved knowledge. Given preserved keys K_0, the null-space projector is computed via SVD on the non-central covariance matrix to reduce computational complexity:

$$\{U, \Lambda, (U)^T\} = \text{SVD}(K_0(K_0)^T)$$

where $\hat{U}$ is the submatrix formed by removing the eigenvectors in U that correspond to non-zero eigenvalues. The projection matrix is defined as $P = \hat{U}(\hat{U})^T$. The sequential editing solution is:

$$\Delta_{\text{AlphaEdit}} = R \cdot K_1^T \cdot P \cdot (K_p K_p^T P + K_1 K_1^T P + I)^{-1}$$

where 1) K_1: The key representing the new knowledge for the current edit and 2) K_p: The keys associated with past edits that must be retained.

This yields an exact guarantee that the model edits occur without interfering with the prior preserved knowledge V_0: $(W + \Delta P)K_0 = WK_0 = V_0$. For implementation, we will use the GitHub repo provided in the paper: https://github.com/jianghoucheng/AlphaEdit

3.2 Policy-space preservation (COPR, adapted)

COPR [7] derives the optimal updated policy as:

$$\pi_t^*(y|x) = \frac{1}{Z_t(x)} \pi_{t-1}(y|x) \cdot \exp\left(\frac{r_t(x, y)}{\beta}\right)$$

where π_{t-1} is the previous policy and r_t encodes the learning signal. Instead of human preference rankings, we will use factual correctness rankings. For each knowledge-update prompt x, we will sample K responses, rank them by ground-truth accuracy, and compute the advantage function $\text{Adv}(x, y_j) = \frac{2j-K-1}{K}$. The training objective combines fitting and regularization:

$$\mathcal{L}_t^{\text{fit}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}_{\text{new}}} \sum_{y \in \mathcal{Y}^x} D_{\text{KL}}\big(P_{y,t}(y|x, \theta) \parallel P_{y,t}^*(y|x)\big)$$

$$\mathcal{L}_t^{\text{reg}}(\theta) = \mathbb{E}_{x \sim \mathcal{R}_{\text{task}}} \sum_{y \in \mathcal{Y}^x} D_{\text{KL}}(P_{y,t}(y|x, \theta) \parallel P_{\text{task}}^*(y|x))$$

where $\mathcal{R}_{\text{task}}$ is a 5% replay buffer.

The key variables are defined as follows:

- $\pi_t^*(y|x)$: The optimal target policy for the current task t.
- $Z_t(x)$: The partition function used for normalization.
- β: The KL-penalty coefficient controlling permitted deviation from the previous policy.
- K: The total number of sampled responses for a given prompt x.
- j: The rank index of response y_j, ordered by factual correctness.
- θ: The trainable parameters of the current policy model.
- $\mathcal{Y}^x$: The set of K sampled responses for prompt x.
- $P_{y,t}(y|x, \theta)$: The re-normalized probability distribution of the current model.
- $P_{y,t}^*(y|x)$: The re-normalized probability distribution under the optimal target policy.
- $P_{\text{task}}^*(y|x)$: The re-normalized optimal probability distribution from a past task, used as a regularization anchor.

4 Approach

4.1 Main task: Financial query decomposition for RAG

We will first fine-tune Qwen2.5-3B on a financial query decomposition task. Given a financial question (such as "How did the Middle East conflict affect energy sector earnings in Q4 2023?"), the model generates n sub-queries $\{s_1, s_2, \ldots, s_n\}$ whose embeddings maximize the Recall@10 of target documents in a fixed retrieval corpus $\mathcal{C}$:

$$\text{maximize} \quad \text{Recall@10}\left(\bigcup_{i=1}^{n} \text{Embed}(s_i), \mathcal{C}\right) \tag{1}$$

For training data, we will use the FNSPID [1] with articles from 2019 to 2021. 2022-2023 data held out as knowledge source updates. To generate the dataset, we will ask questions to a teacher LLM (gpt-4o-mini) and ask it to break down the question into sub-queries for RAG. We will then use the embedding model (Fixed off-the-shelf encoder such as BGE-M3) to embed the sub-queries and vector search to retrieve documents from the FNSPID dataset. The retrieved documents will act as our source of truth. We will generate several sub-queries using the teacher LLM and the top 10 most common documents will be used as the gold retrieval dataset to train the base model. Further, some of these questions will revolve around events with different contexts prior to and after 2022 (such as inflation, presidents, etc). In these cases, we will teacher LLM to generate sub-queries for the based on information pre-2022 and after 2022. The new context needed to answer the questions post 2022 will be used as the knowledge updates. In Table 1, we detail sample questions with context changes post 2022.

Following training, we will then perform knowledge updates based on the 2022-2023 data.

We define knowledge updates as structured financial fact triples $\tau = (s, r, o)$ representing entity changes:

- Leadership changes: $(\text{Apple}, \text{CEO}, \text{Tim Cook}) \rightarrow (\text{Apple}, \text{CEO}, \text{New Person})$
- Acquisitions: $(\text{Company X}, \text{acquired by}, \text{Company Y})$
- Financial metrics: $(\text{Company X}, \text{Q3 revenue}, \$4.2\text{B}) \rightarrow (\$5.1\text{B})$

Every method receives the same inputs derived from each fact triple. The triples can be inputted into AlphaEdit directly, while SFT-based and COPR-adapted methods train on natural-language QA pairs rendered via templates, e.g., $f(\tau) \rightarrow (q_\tau, a_\tau)$. We test at three scales: 200, 1,000 and 3,000 fact edits.

We will perform the knowledge updates using the methods described in table 2. Ideally, when the model learns new facts using either of the two proposed methods, the model's decomposition skill should include these new facts, without degrading prior abilities. Therefore, we will have two tests to check this. First, we will test the Recall@10 on test data (skill + knowledge update requirement). Second, we will test the performance of the model on a prior, unaffected query (skill + prior knowledge requirement). See metrics section for more detail.

4.2 Secondary task: FinQA (time permitting)

We will fine-tune a second Qwen2.5-3B model on FinQA, an established benchmark requiring multi-step numerical reasoning over financial reports (8,281 QA pairs) and then perform knowledge updates. Because FinQA answers are derived from tables already present in the prompt, knowledge updates should not affect performance. If a method degrades FinQA accuracy, it may be causing performance degradation in unrelated capabilities. Therefore, this step serves as a generic forgetting control, and will be included time permitting.

Table 1: Knowledge Update Examples for FNSPID Dataset Generation

Overall Topic	Core Question	Base Knowledge (2019–2021)	Knowledge Update (2022–2023)
Federal Reserve Policy	What is the Federal Reserve's current monetary policy and its stance on interest rates?	Near-zero interest rates and quantitative easing to stimulate the pandemic economy.	Aggressive, historic rate hikes and quantitative tightening to combat inflation.
US Inflation	What is driving US inflation, and how long is it expected to last?	Inflation is "transitory," driven by temporary pandemic supply chain bottlenecks.	The "transitory" narrative is abandoned; inflation hits multi-decade highs due to energy and structural shortages.
Semiconductor Market	What are the primary growth drivers and market outlook for Nvidia (NVDA)?	Revenue driven by gaming GPUs, cryptocurrency mining, and early data center growth.	The Generative AI boom; massive, unprecedented demand for H100 chips for LLM training.
Crypto Market Stability	What is the status of major cryptocurrency exchanges like FTX, and the regulatory environment?	Massive bull market, rapid exchange growth/valuations, and light regulation.	"Crypto Winter," the spectacular collapse and fraud of FTX, and intense SEC regulatory crackdowns.
Regional Banking Health	How stable is the US regional banking sector, and what are their primary risk factors?	Highly stable, flush with pandemic deposits invested in long-term Treasury bonds.	The March 2023 crisis; collapse of SVB and others due to severe duration risk and bank runs.
Twitter Corporate Structure	Who owns Twitter, and what is its corporate structure?	A publicly traded company (TWTR) led by Jack Dorsey and later Parag Agrawal.	Acquired by Elon Musk, taken private, subjected to massive layoffs, and rebranded to "X."
Tech Sector Employment	What are the hiring trends and employment outlook in the major technology sector?	"The Great Resignation," fierce competition for talent, and massive hiring sprees.	"The Year of Efficiency," widespread layoffs across Big Tech, and cost-cutting measures.
Global Energy Markets	How are relations with Russia impacting European and global energy markets?	Standard geopolitical tensions, OPEC+ negotiations, and debates over the Nord Stream 2 pipeline.	Russia invades Ukraine, triggering massive global energy shocks, sanctions, and price spikes.

Method	Type	Description
No update	Lower bound	Task-tuned model, no knowledge injection.
Naive SFT	Baseline	SFT on fact QA pairs, no regularization.
KL-reg SFT	Baseline	SFT + $\lambda \cdot D_{\mathrm{KL}}(\pi_{\mathrm{task}} \| \pi_\theta)$.
Mixed replay	Baseline	SFT on new facts + 5% task data shuffled.
AlphaEdit	Weight-space	Null-space projected fact editing.
COPR-adapted	Policy-space	Advantage-weighted fitting + task replay.
Full retrain	Upper bound	Train from scratch on all data (if compute permits).

Table 2: Summary of experimental methods and baselines.

5 Metrics

We evaluate the models across the following dimensions:

- **Task preservation:** Recall@10 on held-out query decomposition test set.
- **Knowledge absorption:** Accuracy on fact-probe questions about injected triples.
- **Generic forgetting:** FinQA execution accuracy.
- **Locality:** Accuracy on held-out untouched financial facts (stratified by entity/sector)

References

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